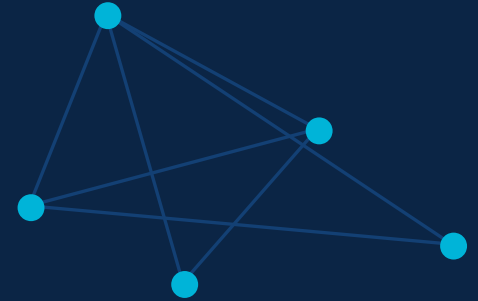




DATA SEMANTICS

Introduction to Data Semantics · The Data Pipeline · Data Cleaning

From meaning in data to clean, trustworthy pipelines

What to cover today

01

Introduction to Data Semantics

Meaning, models and why it matters

02

The Data / Big Data Pipeline

Stages from acquisition to interpretation

03

Data Cleaning

Dirty data, quality dimensions & fixing order

01

Introduction to Data Semantics

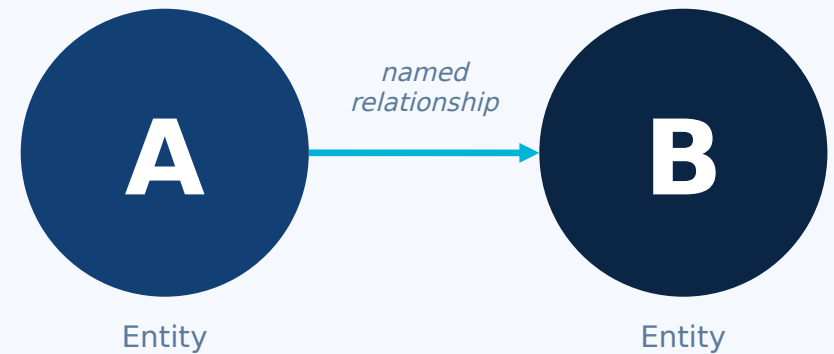
Meaning, models, and why they matter

What Is Data Semantics?

Semantics is the study of meaning — the relationship between signifiers people use (words, phrases, signs, symbols) and the things in the world they denote.

— Panos Alexopoulos, Semantic Modeling for Data (2020)

- “meaning and meaningful use of data” i.e. the effective use of a data object for representing a concept or object in the real world.
- Semantic data = two real objects + one named relationship between them
- Goes beyond storage: it captures what data represents in the real world



Semantic Data = Entity + Relationship + Entity

Why Semantic Modeling Matters

- Matters most when you need to capture more meaning out of your data
- Defines data within the context of the real world and surfaces relationships
- Often called the “brain” of enterprise architecture

Why It Matters

1

Consistency

Ensures consistency and accuracy in analysis

2

Clarity

Reduces ambiguity in interpretation

3

Decisions

Improves data-driven decision-making

Why “100” Alone Means Nothing

100

100 items

Inventory count

100 USD

A monetary value

100 customers

A population size

Semantics is what tells us which one is meant — it clarifies the meaning behind raw values.

Semantics in Practice

1

Banking

Banks use semantic models to integrate customer interactions across channels — delivering one consistent customer experience.

2

Robotic Process Automation

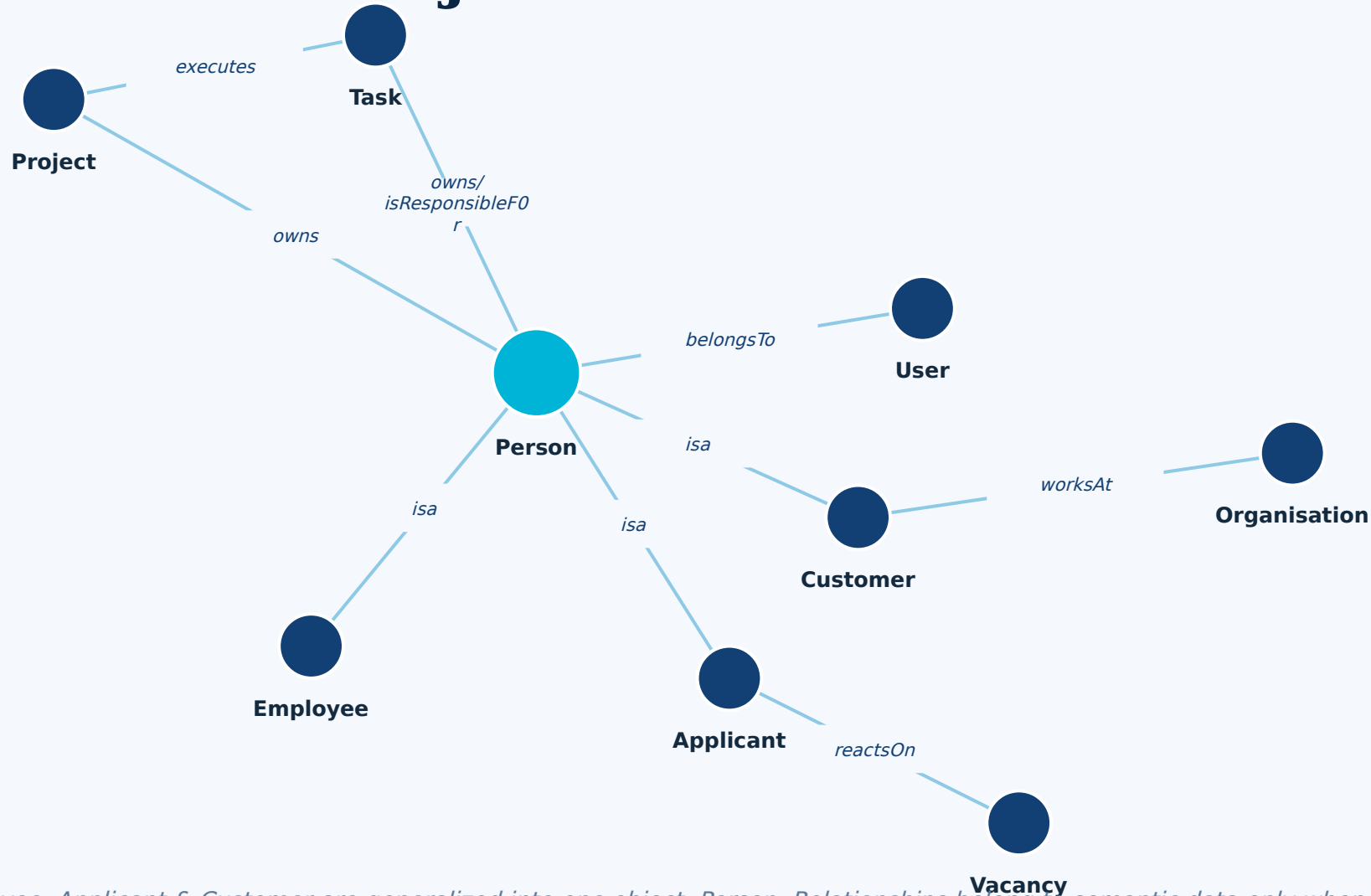
RPA automates simple, repetitive tasks well, but fails to scale on unstructured, complex data with implicit meaning — semantics fills that gap.

3

Search & Recommendation

Knowledge graphs, ontologies and taxonomies power knowledge-driven search, media platforms and social networks.

A Semantic Object Model: “Person”



Employee, Applicant & Customer are generalized into one object: Person. Relationships belong to semantic data only when they are named.

Row & Column Placement Carries Meaning

A piece of data's meaning often comes from where it sits — its row and column position.

Name	Animal	Breed	Sex	Date	Reason for Visit	Notes
Katara	Cat	American Shorthair	F	11/22/23	Checkup	
Grayson	Rabbit	English Lop	M	10/13/23	Yearly Vaccination	
Abby	Dog	German Shorthaired Pointer	F	9/28/23	Appointment	Urinary problems

- “Katara” sits under the “Name” column → we know it's a name, and it belongs to a Cat
- Reading along the row gives breed, visit date, and reason for the visit
- This is implicit semantics: structure (schema) supplies meaning without an explicit label

Modelling Languages for Semantics

- Logic-based data modelling balances tractability with usability
- OWL (Web Ontology Language) — defines ontologies: classes, properties, relationships
- SPARQL — a graph-based query language for querying semantic / RDF data
- Together they enable automated processing, reasoning, search & recommendation

Semantic Data Models



Knowledge Graphs



Ontologies



Taxonomies

Driven by knowledge-driven search, the Semantic Web, social networks & media/streaming platforms.

02

The Data / Big Data Pipeline

From acquisition to interpretation & reporting

What Is a Big Data Pipeline?

- A pipeline coordinates different tasks and technologies to reach a specific data-driven solution
- Big Data breaks traditional schema-first modelling — heterogeneous, distributed, streaming data must be handled as it arrives
- Multiple components must be coordinated to keep data quality and accessibility high for downstream analytics and reporting

Data Semantics Dimensions of a Pipeline

Data

Representing data via schemata or schema-less structures

Metadata

Describing provenance, trust, accuracy & quality

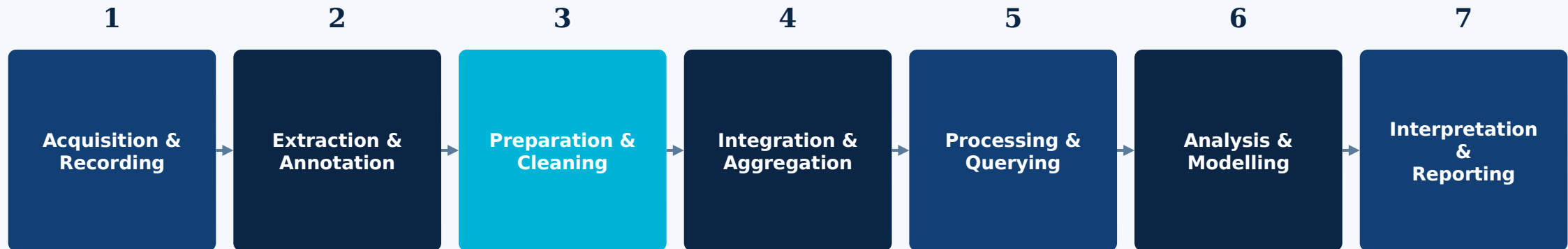
Process

Modelling the pipeline flow itself, end to end

Quality

Task-specific quality checks at every stage

The Seven-Stage Big Data Pipeline



Each stage carries its own data-quality task: source evaluation, metadata extraction, consistency checks, record linkage, quality access, quality of analytics, and task-specific evaluation.

Adapted from Ceravolo et al., “Big Data Semantics”, Fig. 1

Data Acquisition & Recording

- Big Data arises from one or several generating sources that must be interpreted correctly
- Irrelevant data is filtered out before processing begins
- Provenance — tracking how a piece of data was generated — is a major requirement
- The “regulatory barrier” (access, sharing, custody rules) is a key challenge at this stage



Quality Task

Source Quality Evaluation — assessing believability, reliability and reputation of the data source (the Trust cluster).

Data Extraction & Annotation

- Extracts (semi-)structured data from unstructured content: text, images, natural language
- Entities (people, places, companies, events) are identified and linked to a knowledge base
- Multilingual support and domain-specific grammars matter for under-resourced languages
- Notes/SMS/medical records need extra preprocessing due to typos and irregular structure



Quality Task

Quality Metadata Extraction — deriving and annotating metadata that supports later processing stages, e.g. verifying location accuracy against a geographical dictionary.

Data Preparation & Cleaning

- Detects and corrects inaccurate or corrupted records
- Central problem: semantic duplicates — different records representing the same real entity
- Data profiling surfaces functional dependencies and quality problems automatically
- May also increase obfuscation for privacy, IP or strategic-knowledge protection



Quality Task

Consistency Checks, Imputation, Outlier Detection & Duplicate Removal — the four core cleaning sub-tasks.

Data Integration & Aggregation

- Loading data into distributed storage alone is not enough — it must be integrated
- Data Lakes store heterogeneous data in original format; Polystores organize “islands” of data
- A knowledge graph can act as a lightweight mediation layer across sources
- Record linkage identifies which records across sources refer to the same entity



Quality Task

Record Linkage — matching records that represent the same real-world entity across data sources, ideally fully automated at Big Data scale.

Data Processing & Querying

- Hadoop ecosystem: HDFS for scalable, reliable distributed storage
- MapReduce-style engines (Hadoop MapReduce, Spark, Flink) process data in parallel
- SQL-on-Hadoop engines (Hive, Impala, SparkSQL) and SPARQL engines query at scale
- NoSQL databases (key-value, document, graph) extend query options beyond SQL



Quality Task

Quality Access & Quality of Queries — quality-driven data access lets users rely on a source's quality features even when generation is uncontrolled.

Data Analysis & Modelling

- Traditional algorithms assume full-dataset access; streaming data breaks that assumption
- Complex analytics combine ML, data mining, statistics, and mathematics
- Process Mining bridges data mining and Business Process Management using event logs
- Adaptive / runtime models support dynamic, self-adapting large-scale systems



Quality Task

Quality of Analytics — method-dependent metrics (e.g. F-measure) plus method-independent measures such as steady-state, referenceable representations.

Data Interpretation & Reporting



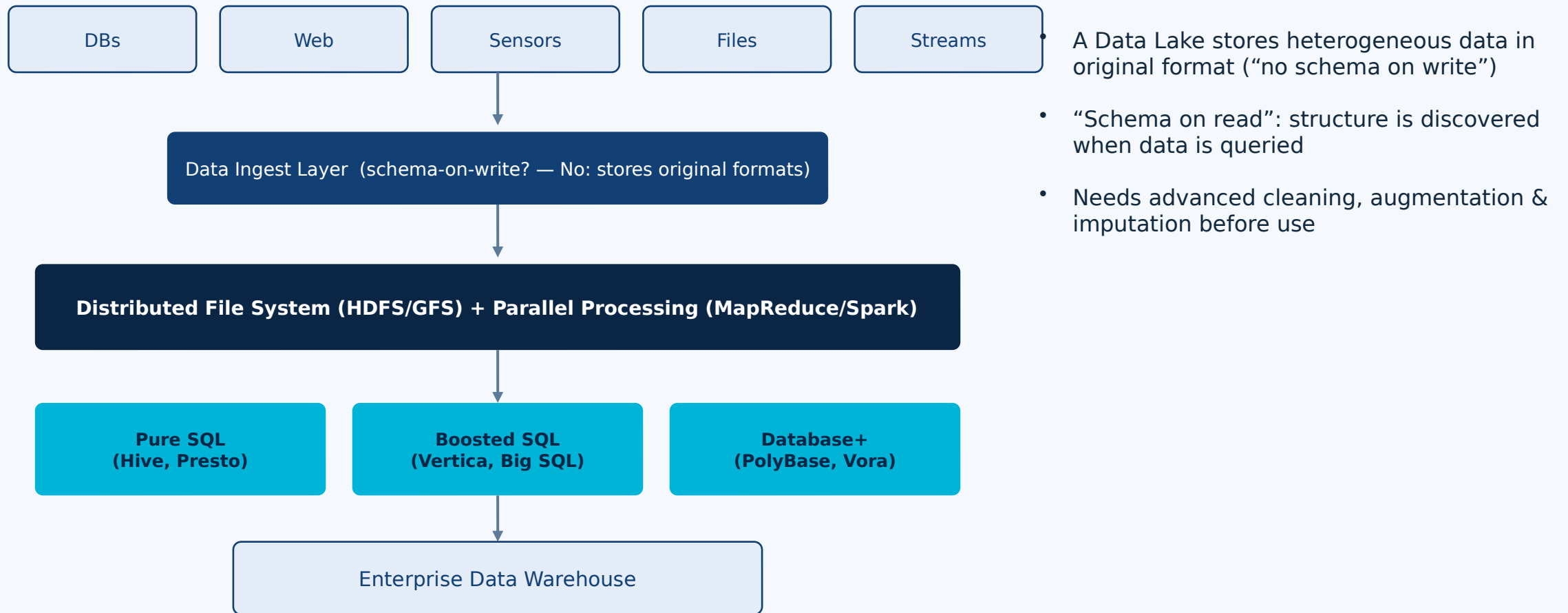
- Visual Analytics combines automatic computation with human visual analysis
- Big Data challenges visualization: volume, velocity and variety all complicate display
- Semantics can drive which visual representation best fits a dataset and task
- Example: Google Flu Trends overestimated cases — a caution on trusting Big-Data models alone



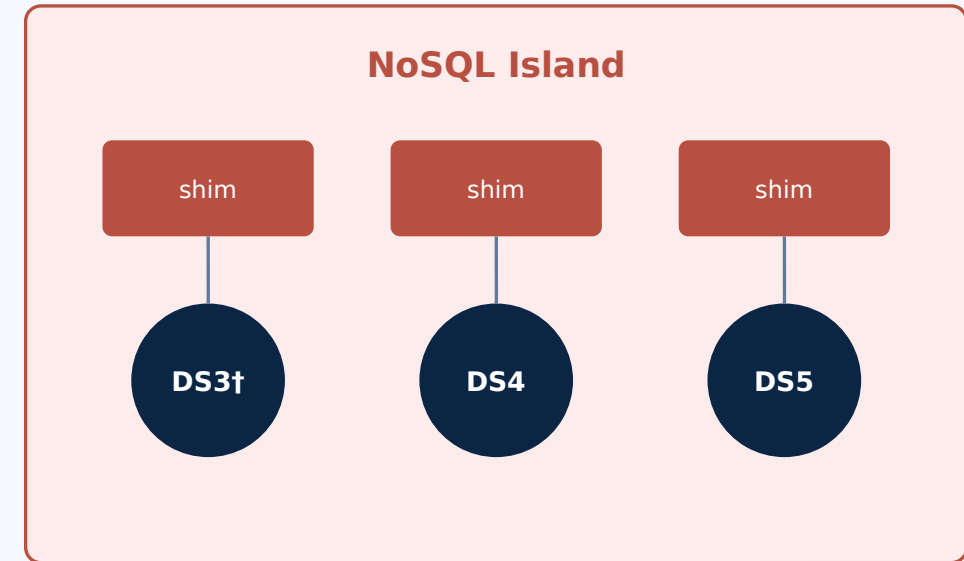
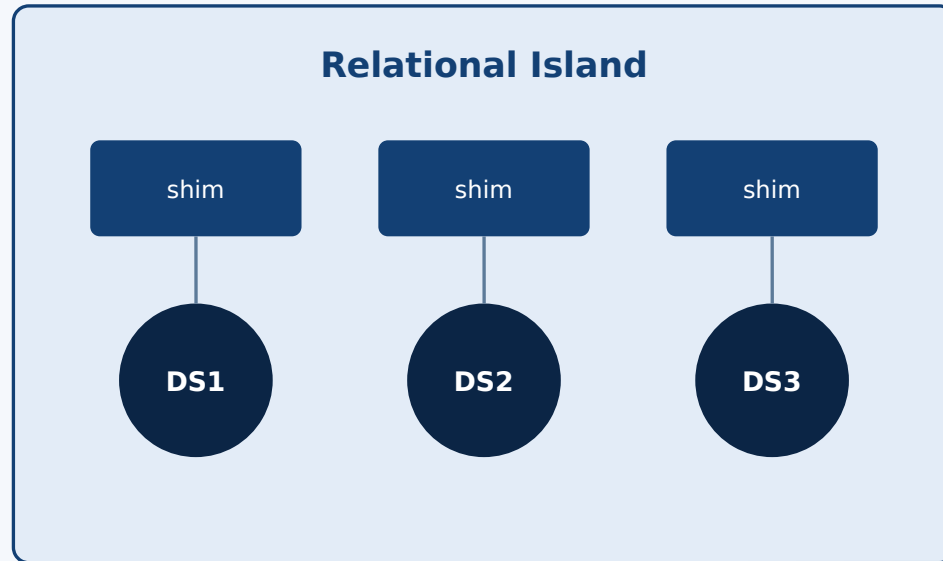
Quality Task

Task-Specific Quality Evaluation — carefully validating the robustness and real-world accuracy of conclusions drawn from Big Data models.

Integration Architecture: The Data Lake



A Lighter Approach: The Polystore



- An island = a collection of storage engines sharing one query language; a “shim” translates island queries into a data source's native language
- DS3 is shared across both islands, so it can be queried from either side
- Query language extended with scope (which island) and cast (output data model) clauses

Guiding Principle: FAIR Data



Findable

Data & metadata can be located easily



Accessible

Retrievable via standard, open protocols



Interoperable

Uses shared vocabularies/formats to integrate with other data



Reusable

Well-described so it can be reused for new purposes

Data Semantics research contributes to Big Data management by pursuing FAIR — Findable, Accessible, Interoperable, Reusable — data & metadata.

03

Data Cleaning

Making data complete, valid, accurate & consistent

What Is Data Cleaning?

- The process of identifying, correcting and removing errors or inconsistencies from data
- Detects and addresses inaccuracies, incompleteness, or irrelevant information within a dataset
- Ensures data is accurate, consistent, and usable for analysis and decision-making
- An essential activity in data preprocessing — it shapes how data is used downstream

“Analysts report spending upwards of 80% of their time on problems in data cleaning.”

— Haas et al., cited in Ceravolo et al.

Why Data Cleaning Is Important

1

Improved Data Quality

Fewer errors, inconsistencies & missing values

2

Better Decision-Making

Insight into comprehensive, current information

3

Increased Efficiency

Clean data is far cheaper to analyze & report on

4

Compliance

Meets regulatory & industry data-quality standards

5

Saves Money

Avoids the cost of errors caused by bad data

6

Builds Trust

Dependable data builds stakeholder confidence

Why Does Data Become Messy?

- 1 Human Error:** Typos, accidental duplicates, wrong values from manual entry/collection
- 2 Multiple Sources:** CRM, inventory, sales systems combine with different formats & labels
- 3 Equipment Errors:** Sensors, meters and gauges malfunction and send faulty readings
- 4 Outdated Information:** Names, addresses, salaries change; stale data becomes untrustworthy
- 5 Software Issues:** Bugs/limits in collection, storage, analysis or visualization tools

Types of Dirty Data



Missing Values

Missing info can produce biased or incomplete conclusions



Duplicate Data

Twofold variations create skewed results



Incorrect Data Types

Wrong-format values (e.g. text in a numeric field) break analysis



Outliers & Anomalies

Unusually high/low values distort statistical results



Inconsistent Formats

Different date formats, capitalization, etc. complicate joins



Spelling / Typos

Misspelled keys get mis-categorized or missed entirely

Data Quality Dimensions



Completeness

Is data missing?

e.g. a blank required field



Validity

Is data invalid?

e.g. negative height, duplicates



Accuracy

Valid, but not accurate

e.g. a person's height — 80 ft



Consistency

Valid & accurate, differently written

e.g. "10KG" vs "10kg"

After the data assessment, categorise each issue found under one of these four dimensions.

The Order of Solving Data Issues



Work through issues in this sequence: fix what's missing first, then messy structure, then invalid values, then inaccuracies, and finally inconsistent formatting.

Cleaning Sub-Tasks in the Big Data Pipeline

1

Consistency Checks

Localize and correct inconsistencies across records

2

Imputation

Localize and correct incomplete / missing data

3

Outlier Detection

Localize values that deviate abnormally from the norm

4

Duplicate / Semantic-Duplicate Removal

Detect records that represent the same real-world entity

Data preparation & cleaning is a dedicated stage of the pipeline — and, at Big Data scale, source- and task-specific (e.g. removing AIS radio-channel glitches in maritime data).

Key Takeaways

1

Data Semantics makes the meaning of data explicit, accurate and shared by humans and machines

2

The Big Data pipeline runs from acquisition through interpretation & reporting, each stage with its own quality task

3

Data Lakes and polystores are common integration architectures for heterogeneous Big Data

4

Clean data — complete, valid, accurate and consistent — is the foundation of trustworthy analytics

5

Solve data issues in order: Completeness → Messy Data → Validity → Accuracy → Consistency

References

- Data Semantics: Cleaning, Aggregating Data, Data Filtration, Data Types (course notes)
- Ceravolo, P., Azzini, A., Angelini, M., et al. “Big Data Semantics.” Journal on Data Semantics (IFIP WG 2.6)
- Wilke, C. O. (2019). Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures (1st Ed.). O'Reilly Media.
- Knaflitz, N. C. (2019). Storytelling with Data: Let's Practice! (1st Ed.). Wiley.
- Wexler, S., Shaffer, J., & Cotgreave, A. (2017). The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios (1st Ed.). Wiley.
- <https://onlinelibrary.wiley.com/doi/book/10.1002/9781119879862>